

# liteOdyssey: A Portable Clinical Reasoning Policy for AI-assisted Rare Disease Diagnosis

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## Abstract

Rare disease diagnosis involves interpreting clinical and genetic findings through complex diagnostic reasoning. We investigated whether this reasoning could be translated into a portable policy for guiding general-purpose large language models (LLMs) without modifying model weights or requiring resource-intensive infrastructure. We developed liteOdyssey, a lightweight framework built through Policy Iteration with Human Feedback, in which clinicians review the model’s reasoning process to iteratively update the policy. This policy guides evidence gathering, tool use, and differential diagnosis generation with an auditable reasoning process. In an external evaluation of 515 Undiagnosed Diseases Network patients, liteOdyssey improved diagnostic accuracy over general-purpose LLMs. These results suggest a strategy for medical AI in which expert reasoning is operationalized as an auditable, reusable policy layer that guides unmodified LLMs without resource-intensive infrastructure.

## 1. Introduction

Rare diseases are individually uncommon but collectively affect an estimated 446 million individuals worldwide<sup>1</sup>. Approximately 80% of known rare diseases have a genetic etiology<sup>2</sup>. Although next-generation sequencing has increased access to genetic evidence, rare diseases remain difficult to diagnose due to limited clinician familiarity, clinical heterogeneity, and symptoms that often overlap with more common conditions. As a result, individuals with rare diseases often experience diagnostic delays averaging five years characterized by repeated evaluations, misdiagnoses, and unnecessary interventions that can contribute to adverse outcomes<sup>3,4,5</sup>. These delays reflect not only limited evidence, but also the challenge of interpreting clinical and genetic findings through a complex diagnostic reasoning process.

Large language models (LLMs) provide new opportunities to support this diagnostic reasoning process. Recent rare disease AI systems improved differential diagnosis by adding structured resources

or infrastructure around LLMs, including knowledge graphs, fine-tuning, solved-case retrieval, and multi-agent orchestration<sup>6,7,8,9,10,11,12,13,14,15,16</sup>. These approaches show that language models can benefit from external biomedical evidence, structured computation, and multi-agent coordination. Some also make model outputs more transparent by producing case-level reasoning traces linked to supporting evidence<sup>12</sup>. While reasoning traces explain what a system did in a particular case, they do not necessarily define the diagnostic strategy the system should follow in future cases. Thus, the central diagnostic reasoning procedure often remains implicit, embedded in model weights, prompts, retrieval choices, or system architecture. When that procedure is not presented as an auditable and reusable policy, it is difficult to inspect, revise, reproduce, or transfer across models and clinical settings.

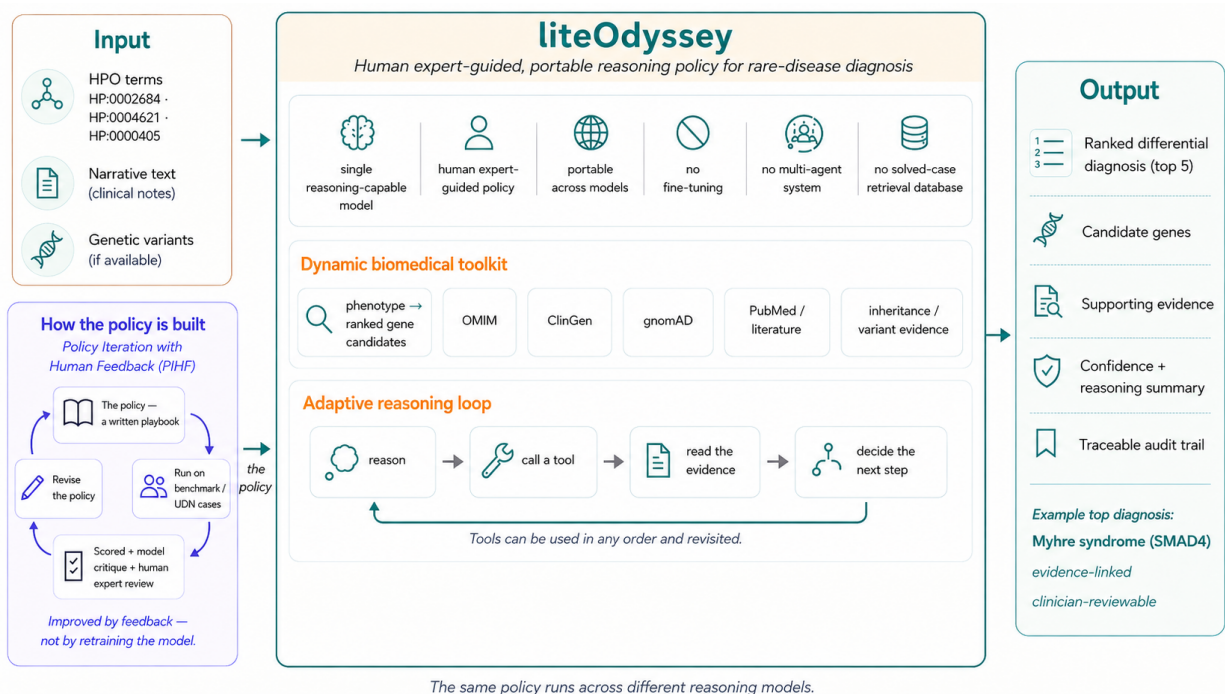
In this paper, we investigated whether expert rare disease diagnostic reasoning could instead be externalized as a portable diagnostic policy that a general-purpose LLM can execute. We developed liteOdyssey, a lightweight, phenotype-first framework for AI-assisted rare disease differential diagnosis (Figure 1). Given a clinical description as input, it automatically extracts structured phenotypes as Human Phenotype Ontology terms<sup>17</sup>, queries public biomedical knowledge sources, generates and refines disease and gene hypotheses, and returns a ranked differential diagnosis with a transparent, step-by-step account of the evidence and reasoning behind it. At the center of the system is a structured diagnostic policy that specifies how clinical and genetic evidence should be gathered, evaluated, and incorporated into diagnostic ranking. The policy is clinician-readable, model-executable, and supported by biomedical tools that provide deterministic, provenance-carrying access to public phenotype, gene, disease, inheritance, and literature knowledge.

To build this policy, we introduce Policy Iteration with Human Feedback (PIHF), a weight-free optimization process in which clinicians review the model’s reasoning process, identify recurrent diagnostic errors, and convert those errors into policy updates. Rather than changing model weights, PIHF improves the diagnostic procedure the model follows. The resulting system operationalizes rare disease diagnosis as a policy-guided reasoning process, supported by source-traceable biomedical tools and executable without fine-tuning, custom model training, large external case banks, or heavy computing infrastructure.

We evaluated liteOdyssey across curated public benchmarks, cumulative ablations, cross-model experiments, and a real-world cohort from the Undiagnosed Diseases Network. Public benchmarks comprising more than 1,200 cases across 722 rare diseases tested whether policy-guided reasoning improved disease and gene ranking from structured clinical features. Ablations separated the contribution of the policy from prompting and access to the same biomedical tools, while cross-model experiments tested whether the same policy could be executed by closed and open-weight models without fine-tuning. The real-world Undiagnosed Diseases Network cohort tested whether an adapted policy retained performance gains in 515 patients whose diagnoses were not established after standard clinical evaluation. Across these settings, liteOdyssey improved ranked disease and gene differentials over matched language model baselines, with larger gains in curated public benchmarks and smaller but statistically significant gains in the clinical cohort.

These results show that expert diagnostic reasoning can be operationalized as an external policy that is auditable, updatable, and reusable across models. liteOdyssey does not require training a new diagnostic model or replacing existing AI architectures. Instead, it defines a reproducible reasoning layer that can be inspected before deployment, audited after each case, revised as medical knowledge changes, and reused across model families. For rare disease diagnosis, where evidence is often incomplete and missed diagnoses can prolong the diagnostic odyssey, liteOdyssey suggests that medical AI can improve not only through larger models or heavier infrastructure, but also

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b

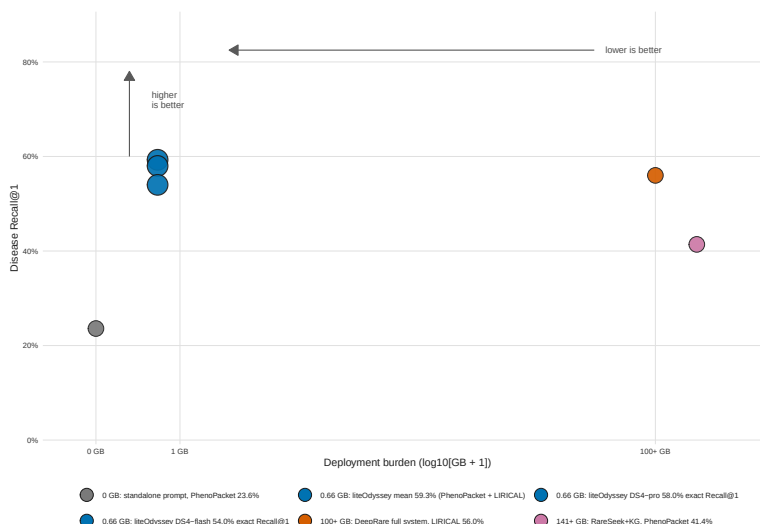


Figure 1. Overview of liteOdyssey and performance relative to system size. (A) liteOdyssey guides a single reasoning-capable LLM through phenotype interpretation, biomedical tool use, evidence review, and ranked differential diagnosis using a human expert-guided diagnostic policy. The policy is developed through PIHF, which uses benchmark performance, the model’s reasoning process, model critique, and clinician review to revise the written policy rather than retrain the model. (B) Disease Recall@1, defined as the proportion of cases in which the correct diagnosis was ranked first, is plotted against approximate local system size, measured by the amount of user-side data or infrastructure required to run each system. liteOdyssey achieved high accuracy with limited local resources.

through explicit, portable representations of expert reasoning that make diagnostic workflow more transparent, reproducible, and clinically accountable.

## 2. Results

### **Policy Iteration with Human Feedback (PIHF) externalizes diagnostic reasoning as an AI-executable policy**

To determine whether expert rare disease reasoning can be represented outside an LLM as a reusable policy, we developed Policy Iteration with Human Feedback (PIHF), a weight-free process in which a frozen LLM runs a candidate policy, clinicians review the model’s reasoning process, identify recurrent diagnostic errors, and revise the policy rather than modify internal model weights. The resulting liteOdyssey policy specifies the reasoning process beginning from a patient’s clinical features to a ranked differential diagnosis, including which evidence to gather, how to weigh competing hypotheses, and when to dynamically call biomedical tools (Figure 1). Thus, the optimized object of PIHF is not a trained model or a one-time prompt, but a clinician-readable, model-executable, external reasoning layer that can be audited and updated.

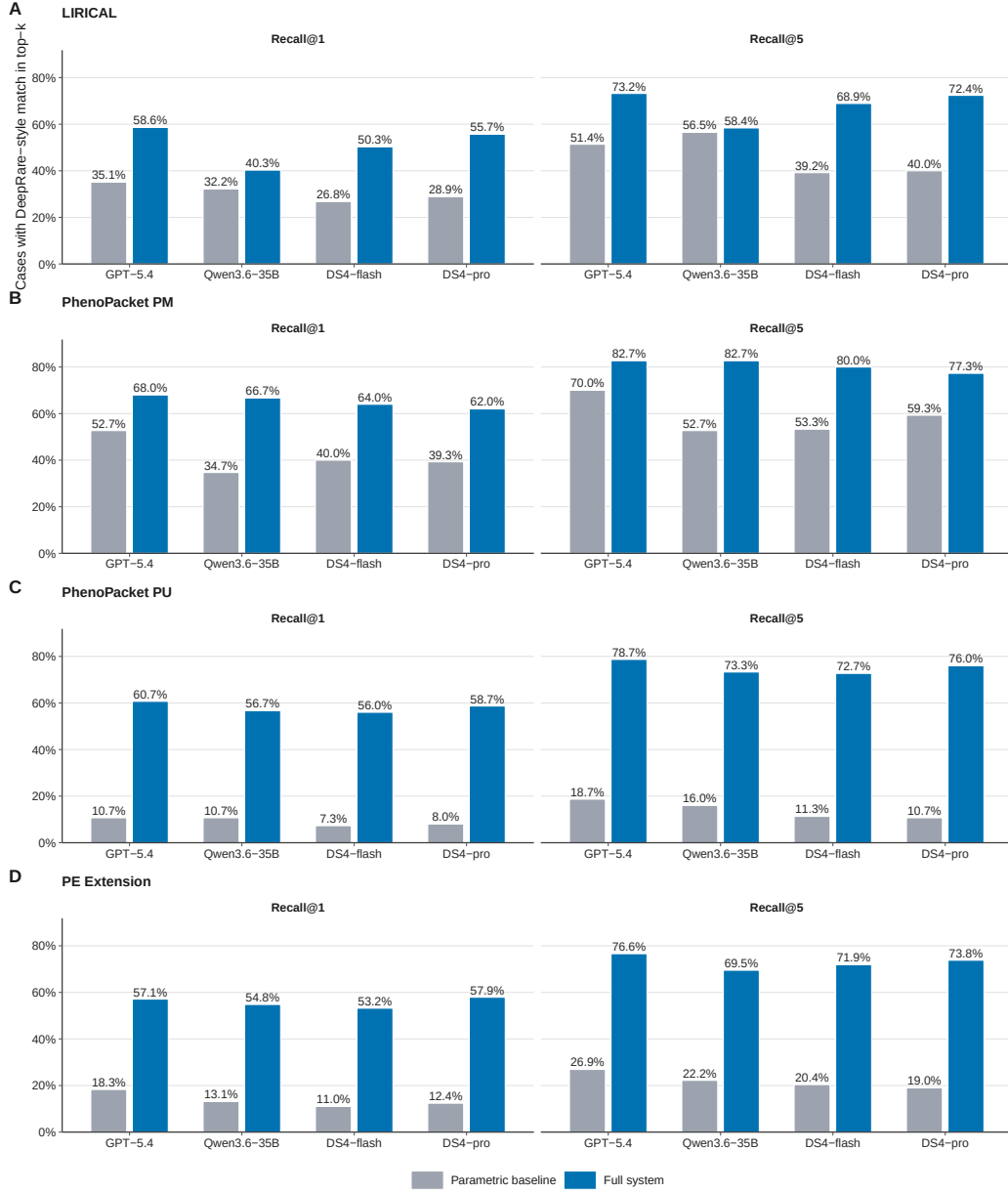
liteOdyssey applies this policy to phenotype-first rare disease diagnosis. Given clinical information such as Human Phenotype Ontology (HPO) terms or narrative text, the system uses a single reasoning-capable LLM to execute the policy, call deterministic biomedical tools, integrate retrieved evidence, and return a ranked differential diagnosis with candidate genes, supporting evidence, confidence assessment, and an auditable case-level reasoning process. liteOdyssey requires no fine-tuning, multi-agent architecture, solved-case retrieval database, or knowledge graphs. Its local resource requirements are therefore small relative to systems that rely on heavier external infrastructure, while preserving a transparent policy and reasoning process that can be inspected and revised.

### **A portable policy improves rare disease diagnosis on public benchmarks and transfers across models**

If PIHF had captured useful diagnostic reasoning, then supplying the resulting policy to general-purpose language models of different sizes should improve diagnosis from the same clinical features. We tested this on 1,243 public rare disease cases from LIRICAL and the PhenoPacket Store. Each case provided clinical features, and the task was to rank the causal disease within a differential diagnosis.

Using GPT-5.4, liteOdyssey ranked the correct disease first (Recall@1) in 58.6% of LIRICAL cases, compared with 35.1% for the parametric baseline (i.e., off-the-shelf GPT-5.4), and in 59.6% of PhenoPacket Store cases, compared with 22.9% for the baseline (Figure 2). liteOdyssey also achieved higher Recall@5 than the baseline on LIRICAL (73.2% vs. 51.4%) and the PhenoPacket Store (78.0% vs. 32.9%). Across all 1,243 public cases, disease Recall@1 was 59.3% for liteOdyssey and 26.6% for the baseline.

The largest performance gain was observed in the subset of PhenoPacket Store cases in which the diagnoses could not be mapped to Orphanet disease identifiers. In this more difficult subset, Recall@1 increased from 10.7% with the parametric baseline to 60.7% with liteOdyssey, and Recall@5 increased from 18.7% to 78.7%. This pattern suggests that the policy was most useful



GPT-5.4/Qwen bars use DeepRare-style scoring; DS4 paired bars use independent Sonnet Prompt-2 scoring; exact disease-ID is retained as a stricter sensitivity check.

Figure 2. Policy-guided reasoning improves disease ranking across public rare disease benchmarks. Disease Recall@1 and Recall@5 are shown for the parametric baseline and the full liteOdyssey system across four public benchmark settings: (A) LIRICAL, (B) PhenoPacket mapped cases, (C) PhenoPacket unmapped cases, and (D) PhenoPacket extension cases. Recall@1 is defined as the proportion of cases in which the correct diagnosis was ranked first; Recall@5 is the proportion in which the correct diagnosis appeared within the top five ranked candidates. The parametric baseline used the same language model to diagnose from the same clinical features without the liteOdyssey diagnostic policy or biomedical tools. The full system used the PIHF-derived policy-guided workflow with biomedical tool use. Results are shown across GPT-5.4, Qwen3.6-35B, DeepSeek-V4-Flash-Preview, and DeepSeek-V4-Pro-Preview. GPT-5.4 and Qwen3.6 outputs were scored with the DeepRare-style judge; DeepSeek-V4 outputs were scored with an independent Claude Sonnet judge, with exact disease-ID matching retained as a stricter sensitivity check.

when diagnosis could not rely on direct recognition of a well-indexed disease label, but instead required structured diagnostic reasoning. On the 370-case LIRICAL benchmark, liteOdyssey also exceeded the closest published reasoning-based comparator, DeepRare, which reported 39.5% Recall@1 without retrieval from a solved-case database and 51.6-56.0% with retrieval from 67,795 solved cases. liteOdyssey reached 58.6% on the same benchmark using a single reasoning model, public biomedical tools, and no solved-case database.

The same policy transferred across model families. Developed on a single closed model (GPT-5.4), the policy was then executed unchanged on other general-purpose reasoning models spanning a range of sizes, including the open-weight Mixture-of-Experts models Qwen3.6-35B-A3B (35 billion total parameters, 3 billion active), DeepSeek-V4-Flash-Preview (13 billion active), and DeepSeek-V4-Pro-Preview (49 billion active). Across LIRICAL and PhenoPacket Store subsets, the policy improved each model over its own parametric baseline (Figure 2). Smaller models improved substantially with the policy, and larger open-weight models approached the performance of the frontier closed model under the same policy, consistent with scaling laws<sup>18,19</sup>. These results indicate that the policy is not a model-specific prompt but a portable reasoning layer reusable by different models.

### **Performance gain is attributable to policy-guided reasoning, not prompting or resource access alone**

To determine whether liteOdyssey’s performance gain could be explained by prompt engineering or by access to biomedical resources alone, we conducted a series of ablation analyses using the same GPT-5.4 model and exact OMIM disease identifier matching on LIRICAL cases. The base model prompted only with clinical features achieved 34.3% Recall@1. An optimized diagnostic prompt, generated through iterative critique and revision but without external tools, improved performance modestly to 36.8%. Providing that optimized prompt with access to the same public biomedical resources used by liteOdyssey increased Recall@1 to 44.9%. The full liteOdyssey system, which guided the model with a diagnostic workflow and gave it access to the specialised-engineered tools within that workflow, reached 55.9% Recall@1. Recall@5 showed the same pattern, with 48.6%, 52.7%, 61.9%, and 72.7% for the base prompt, optimized diagnostic prompt, optimized diagnostic prompt with access to biomedical resources, and the full liteOdyssey system, respectively.

These results separate three increasingly structured forms of model guidance. A better prompt produced only a small improvement in diagnostic accuracy. Access to biomedical resources produced an intermediate improvement. The largest performance gain was observed when the model followed the PIHF-derived reasoning workflow. Therefore, the policy contributes beyond giving the model more instructions or more information; it changes how the model gathers, weighs, and revisits evidence before producing a differential diagnosis.

### **PIHF adapts the policy to a real-world cohort of diagnostically complex patients**

While public benchmarks serve as useful controlled tests, they do not necessarily capture the complexity of patients whose diagnoses remain unresolved after extensive prior workup. We evaluated liteOdyssey on 515 patients from the Undiagnosed Diseases Network (UDN), spanning 447 distinct diagnoses across 15 U.S. clinical sites. These patients had rare, diagnostically complex conditions that were not resolved through prior clinical workup and were later diagnosed through a UDN evaluation.

For the UDN cohort, we further applied the same PIHF process to extend the policy developed on the public benchmarks, adding targeted reasoning steps for the atypical and multi-system presentations these patients often show. On the full UDN cohort, liteOdyssey achieved a Recall@1 of 20.4%, compared with 16.7% for the parametric baseline (+3.7 percentage points; paired exact McNemar  $p = 0.027$ ; 95% CI, 0.6 to 6.8), and a Recall@5 of 31.8% vs. 25.6% on the baseline (+6.2 points;  $p = 7 \times 10^{-4}$ ) (Figure 3).

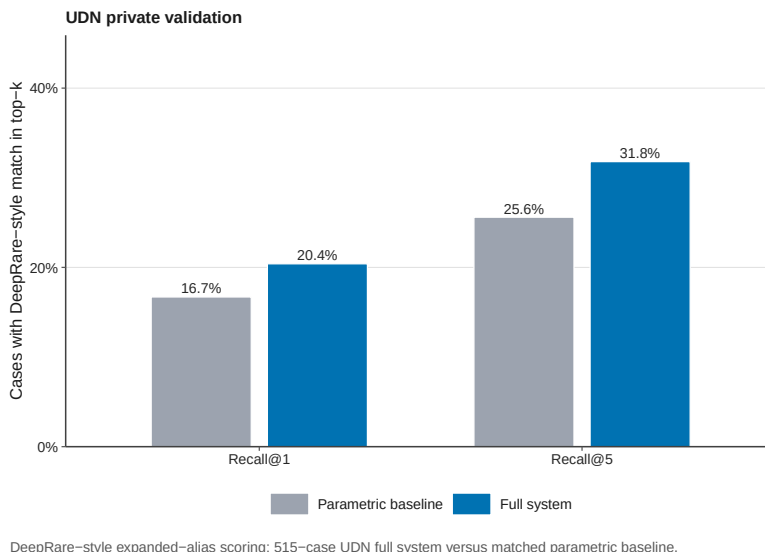


Figure 3. Policy-guided reasoning improves diagnosis in a real-world UDN cohort. Disease Recall@1 and Recall@5 are shown for 515 patients evaluated by the Undiagnosed Diseases Network. The full liteOdyssey system is compared with a matched parametric baseline using the same language model and clinical features but without the PIHF-derived diagnostic policy or biomedical tools. liteOdyssey increased Recall@1 from 16.7% to 20.4% and Recall@5 from 25.6% to 31.8%. Recall was scored using DeepRare-style expanded-alias matching.

Absolute differential diagnostic accuracy was lower on the UDN cohort than public benchmarks, which was expected as the former is enriched for patients with diagnostically complex presentations. However, liteOdyssey still showed measurable performance gains in the UDN, and the magnitude of improvement was similar in the full and hold-out UDN cohorts. These results suggest that PIHF can adapt a diagnostic policy developed on curated public benchmarks to a diagnostically challenging, real-world patient cohort with incomplete genetic information, demonstrating the policy’s portability to new clinical contexts.

To test whether these LLM-judge gains held up under expert clinical review, two physicians independently adjudicated a prespecified 120-case validation sample, blinded to model identity, scoring each output on the Bond differential-diagnosis scale<sup>20</sup>; a third blinded physician resolved the 66 output-level disagreements. On physician-adjudicated scores weighted to the full 515-case cohort, liteOdyssey scored higher than the parametric baseline (weighted mean 3.15 vs. 2.91; paired difference +0.24, 95% CI +0.09 to +0.40; two-sided  $p = 0.002$ ). Physicians recorded the exact diagnosis for 31.8% of liteOdyssey outputs versus 26.6% for the baseline, exact or very close matches for 41.1% versus 35.0%, and complete misses for 12.7% versus 15.9% (Figure 4); inter-rater agreement was high (quadratic-weighted  $\kappa = 0.909$ ). Physician agreement with the LLM judge was strongest on high-confidence outputs (86–87%) and lower on moderate- and low-confidence outputs, indicating

that the judge’s confidence functions as a calibration signal.

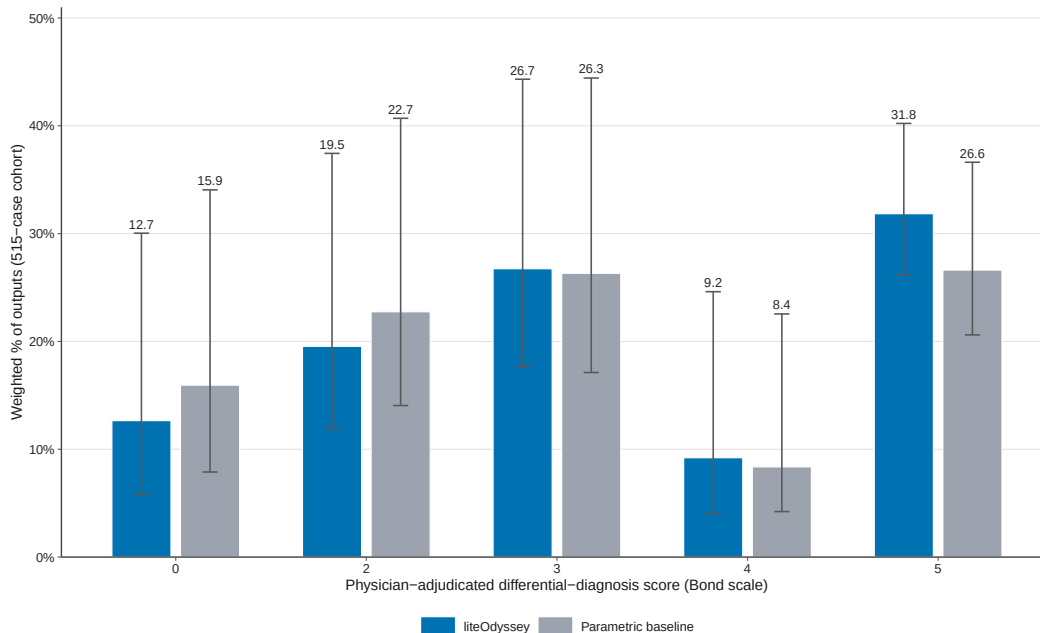


Figure 4. Blinded physician review of the UDN cohort corroborates the policy gain. Distribution of physician-adjudicated differential-diagnosis scores (Bond scale: 0, no close suggestion; 2, related but unlikely helpful; 3, closely related; 4, very close; 5, exact diagnosis) for liteOdyssey and the matched parametric baseline, weighted to the full 515-case Undiagnosed Diseases Network cohort. Two physicians blinded to model identity scored a prespecified 120-case sample and a third physician adjudicated their disagreements. liteOdyssey scored the exact diagnosis (5) in 31.8% of outputs versus 26.6% for the baseline and had fewer complete misses (0) at 12.7% versus 15.9%; the weighted mean score was higher for liteOdyssey (3.15 versus 2.91; paired difference +0.24, 95% CI +0.09 to +0.40). Error bars are pointwise 95% Wilson intervals.

## The policy generalizes to development-excluded cases

Because PIHF produces an external policy rather than model-specific weights, we then evaluated whether the same policy works beyond the development cases used to produce it. For the public benchmarks, policy development used only 50 LIRICAL cases and no PhenoPacket Store cases. The independent hold-out set therefore included the remaining 320 LIRICAL cases and all 873 PhenoPacket Store cases. The 1,193 hold-out cases comprised 679 distinct diseases not represented during policy development.

Performance gains persisted on the independent hold-out cases. On the 320 hold-out LIRICAL cases, liteOdyssey achieved 57.5% Recall@1, compared with 34.1% for the parametric baseline, and 71.9% Recall@5, compared with 50.9% for the baseline (Figure 5). These performance gains were nearly identical to those observed in the full LIRICAL cohort: +23.4 versus +23.5 percentage points for Recall@1, and +20.9 versus +21.9 points for Recall@5. In the UDN cohort, where 50 cases were used for policy development and 465 were held out, hold-out Recall@1 improved from 15.5% to 19.1% (paired exact McNemar  $p = 0.036$ ), and hold-out Recall@5 improved from 23.7% to 29.9% ( $p = 0.0015$ ). liteOdyssey’s performance improvement was the same in the full and hold-out UDN cohorts: +3.7 points for Recall@1 and +6.2 points for Recall@5 (Figure 5). These results

argue against case-level overfitting and support the interpretation that PIHF improved a reusable diagnostic procedure.

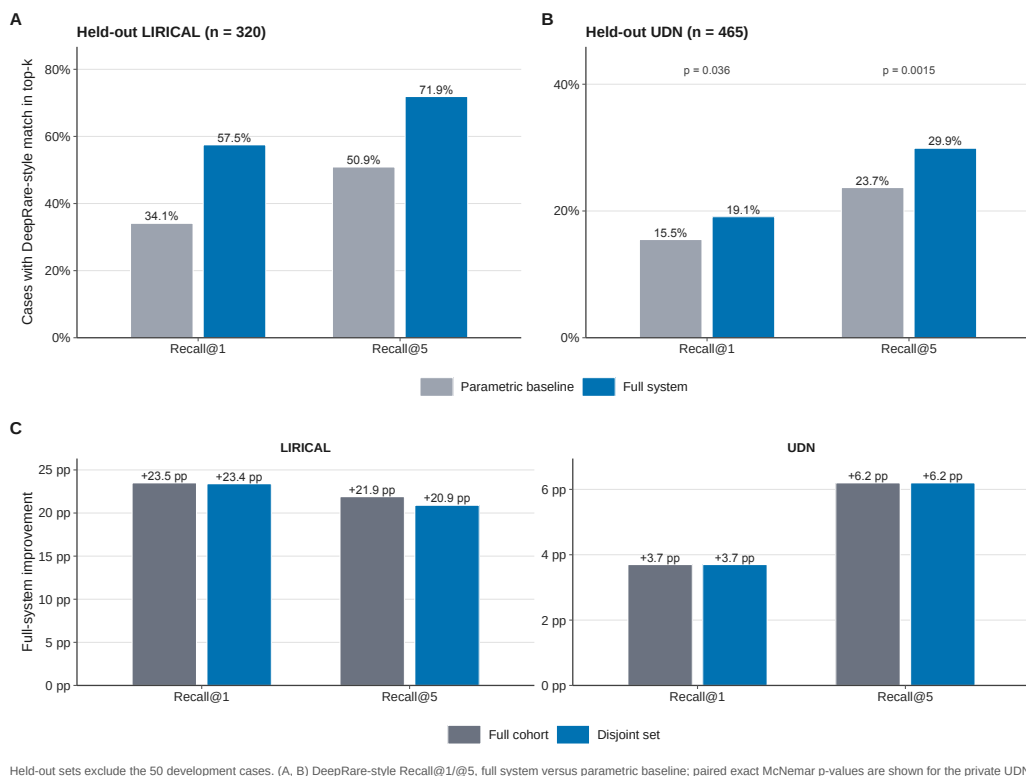


Figure 5. Policy gains persist on development-excluded cases. Held-out analyses exclude the 50 cases used during policy development. (A) Disease Recall@1 and Recall@5 on 320 development-excluded LIRICAL cases. (B) Disease Recall@1 and Recall@5 on 465 development-excluded Undiagnosed Diseases Network (UDN) cases. Bars compare the matched parametric baseline with the full liteOdyssey system. Paired exact McNemar p-values are shown for the private UDN cohort. (C) Full-system-minus-baseline improvements for the full cohort and development-excluded set, showing stable gains across the disjoint split for both LIRICAL and UDN. Recall was scored using DeepRare-style expanded-alias matching.

## Policy-guided outputs are transparent and auditable

liteOdyssey returns not only a ranked differential diagnosis, but also the reasoning process that produced it. For each case, the reasoning process lists the clinical features considered, genes and diseases evaluated, biomedical resources queried, evidence supporting or weakening each candidate hypothesis, and rationale for the final ranking. This reasoning process does not explain the model's internal computations. It provides a clinical audit trail showing what evidence was gathered and how it influenced the differential.

Worked cases show how policy-guided reasoning differs from parametric reasoning in off-the-shelf LLMs. In one LIRICAL case involving Myhre syndrome, the parametric baseline matched the patient's skeletal and sclerosing features to unrelated skeletal dysplasias and did not rank the correct diagnosis highly. In comparison, liteOdyssey connected the patient's phenotypes to the gene *SMAD4*, identified Myhre syndrome through disease lookup, checked the strength of the gene-

disease relationship, and kept the more specific diagnosis as the top-ranked candidate after reflective review. Reviewing the reasoning in other cases revealed additional diagnostic errors, such as early anchoring on the wrong diagnosis, and instances where structured search made the explanation more complicated than necessary. This ability to inspect the reasoning process is central to the policy-based design, which not only improves diagnostic accuracy, but also makes the diagnostic process legible for clinicians to examine, audit, and update.

### 3. Discussion

We showed that expert reasoning for rare disease diagnosis can be externalized as an auditable policy that improves existing language models across public benchmarks, hold-out cases, model families, and a diagnostically complex cohort from the UDN. The central advance is not only the performance of liteOdyssey, but the method used to produce it. Through PIHF, expert review of model reasoning failures was converted into revisions of a persistent diagnostic policy. This policy specifies how clinical features should be interpreted, how biomedical evidence should be queried, how competing hypotheses should be compared, and when an early diagnostic hypothesis should be reconsidered before final ranking. Thus, PIHF makes the reasoning procedure itself an object of optimization.

This distinction has important implications for medical AI. In reinforcement learning with human feedback, feedback is used to update model parameters. In PIHF, feedback updates a clinician-readable, model-executable policy. The policy can be reviewed, versioned, and revised without retraining the underlying model. This is particularly important in clinical domains, where transparency of AI reasoning procedures is critical and where medical knowledge, standards of evidence, and local practice may change over time. PIHF therefore offers a complementary approach to infrastructure scaling and fine-tuning. It improves model behavior by modifying the reasoning process available at inference, while preserving that process as an external policy. This places PIHF among weight-free, in-context approaches to improving model behavior, but with the distinct goal of producing an auditable clinical reasoning policy<sup>21,22,23,24</sup>.

The empirical results support this interpretation. liteOdyssey improved disease ranking on two phenotype-first public benchmarks and demonstrated the largest performance gains in the subset of PhenoPacket Store cases in which diagnoses could not be mapped to Orphanet identifiers. This finding suggests that the policy was most useful when diagnosis required structured evidence gathering rather than direct recognition of a familiar disease label. A cumulative ablation further separated the contribution of the policy-guided workflow from simpler alternatives. Prompt optimization produced only a modest improvement, access to biomedical resources produced an intermediate improvement, and the largest performance gain was observed when the model followed the PIHF-derived diagnostic policy. Because the model and input features were held fixed, these comparisons support the conclusion that the gain arose from the combined effect of the policy-guided procedure and its tools. The model was not merely given more information. It was given a structured method for gathering, evaluating, and synthesizing evidence before generating a ranked differential diagnosis. Beyond disease ranking, the same policy improved causal-gene ranking as a secondary output, with a +37.6-point Recall@1 gain on the PhenoPacket Store and a +2.1-point gain on the UDN cohort (Supplementary Information).

The policy also showed evidence of portability. Performance gains were observed on hold-out cases excluded from policy development: of the 722 diseases spanned by the public benchmarks, 679 never appeared during development, and 407 of the 447 UDN diagnoses were held out (Appendix

Table A2). The same policy improved multiple reasoning-capable models, including open-weight models. This finding suggests that the policy is not simply a prompt adapted to one model family. In the UDN cohort, absolute differential diagnostic accuracy was lower than in public benchmarks, as expected for patients whose diagnoses remained unresolved after standard evaluation and whose genetic findings were not available to the system. Even so, applying the same PIHF process to extend the policy produced statistically significant gains that were stable in the full and development-excluded UDN cohorts. These results show that the same PIHF process can evolve an explicit reasoning policy to meet a diagnostically challenging clinical setting while achieving measurable performance gains.

Our PIHF approach also has implications for reframing the relationship between clinicians and AI systems. Many clinical AI workflows place the clinician at the end of the pipeline, where the main task is to interpret, accept, or reject a model output. PIHF places clinicians earlier in the process, where they help define and revise the reasoning procedure that future model runs will follow. The case-level reasoning process produced by liteOdyssey is not a mechanistic explanation of the model’s internal computations. It is a process audit that records which features initiated the search, which tools were used, which evidence supported or weakened each candidate, and why the final ranking was produced. This form of auditability gives clinicians a transparent record of the diagnostic process that can be examined, challenged, and improved.

The lightweight structure of liteOdyssey may also be relevant for clinical implementation. The system does not require fine-tuning, a multi-agent architecture, a large solved-case retrieval database, or a locally maintained knowledge graph. It uses a single reasoning-capable model, a written policy, and deterministic tools over public or cached knowledge sources. This design is not a substitute for comprehensive clinical systems that incorporate genomic, laboratory, biochemical, imaging, longitudinal, and environmental information. Rare disease diagnosis will likely benefit from both explicit reasoning policies and richer clinical infrastructure. The present study instead identifies a complementary design principle for medical AI. When the limiting factor is how evidence is gathered and weighed, expert reasoning can be made explicit as a policy layer rather than left implicit in model weights or distributed across opaque system components.

Several limitations should be considered. This study evaluated retrospective disease ranking against known diagnoses, rather than prospectively assessing the impact of model use on clinical decision-making or patient outcomes. Nevertheless, in blinded physician adjudication liteOdyssey’s differentials were more likely than the baseline’s to contain the exact diagnosis and less likely to be judged unhelpful (Figure 4); reviewers also more often noted that its differential could change the patient’s subsequent diagnostic workup, an informal observation on the adjudicated cases rather than a prespecified analysis. Together these suggest clinical value beyond ranking metrics alone. liteOdyssey was evaluated primarily using phenotype representations and did not systematically incorporate laboratory, biochemical, imaging, or variant-level evidence, which are essential for many metabolic, mitochondrial, and molecularly defined disorders. The benchmarks consisted of monogenic rare disease cases, so performance on polygenic rare disease and blended phenotypes was not assessed. Finally, the reasoning process documents the policy-guided evidence path. It should not be interpreted as direct access to the internal computations of the language model.

Future work should evaluate policy-guided reasoning prospectively and assess its clinical impact, including effects on safety, efficiency, cost, clinician trust, and patient outcomes. Such studies should also determine which elements of diagnostic policy generalize across tasks and which require local adaptation. Future work should also test whether a stabilized PIHF policy can be distilled into model weights, trading some legibility and portability for lower inference cost, building on

context-distillation and adapter-generation methods<sup>25,26,27</sup>. More broadly, our findings suggest that progress in medical AI may depend not only on scaling models, but also on making expert reasoning explicit, portable, and auditable.

## 4. Methods

### 4.1 System overview and diagnostic policy

liteOdyssey is a phenotype-first reasoning framework for diagnosing rare genetic diseases. It takes input patient clinical features (either in free text or encoded as Human Phenotype Ontology [HPO] terms) and outputs a ranked differential diagnosis with candidate diseases, associated genes, supporting evidence, and a traceable reasoning summary. liteOdyssey has three core components: an external natural-language diagnostic policy, a biomedical tool library that provides structured access to curated public knowledge sources, and a reasoning-capable language model that executes the policy end-to-end. The policy specifies how clinical features are interpreted, which biomedical tools from the library (§4.2) are available to use, how to weigh the evidence they return, and when to revisit an earlier step — so the tool library and the reasoning workflow are not separate modules but components of a single policy artifact. Because the policy is an external artifact rather than fine-tuned weights — requiring no model-weight updates, solved-case retrieval, or multi-agent orchestration — the same specification can be run across model families (see Results), and a clinician can inspect, contest, or revise it without retraining a model. As it executes, the model integrates evidence across tool calls and produces a continuous case-level reasoning process.

The policy developed by PIHF (§4.3) is an eight-phase reasoning workflow: (0) pattern recognition, (1) phenotype-based candidate gene generation, (2) evidence triage, (3) deep disease-level investigation, (4) confidence assessment, (5) corrective search, (6) reflective adjudication, and (7) a ranked final output. The workflow is structured but not strictly linear: the system can revisit earlier phases when new evidence changes the working differential or when uncertainty remains high, and each phase specifies which biomedical tools (§4.2) it draws on. The full phase-by-phase specification is provided in Appendix B.

### 4.2 Biomedical tool library

liteOdyssey was built with a broad toolkit for genetic diagnosis. Because this study evaluates the phenotype-first benchmarks, we describe here the subset of tools those evaluations exercised; each is designed to approximate an external knowledge source consulted during rare-disease evaluation. Each tool addresses a focused clinical question, and no individual tool returns a final differential diagnosis. Each is engineered for the diagnostic task rather than for general search: it returns deterministic, source-traceable evidence for that question, so a second run reproduces the same result and a clinician can trace each piece of evidence to its public source. The language model is responsible for interpreting the returned evidence from each tool and integrating it into a ranked differential.

This evaluated subset comprises five knowledge tools: phenotype-to-gene ranking (Monarch Initiative<sup>28</sup>); gene-to-disease and disease-family lookup with inheritance and onset (OMIM<sup>29</sup>); gene-disease validity and dosage sensitivity (ClinGen<sup>30</sup>); gene constraint (gnomAD<sup>31</sup>, combined with OMIM); and literature retrieval (PubMed). The broader toolkit also includes variant-mode components — variant quality filtering and inheritance-pattern analysis — that activate when sequencing

or trio data accompany a case, extending the same framework from phenotype-only input toward variant-informed diagnosis; these were not exercised on the phenotype-only public benchmarks and are not evaluated in this study. These five tools — the clinical question each answers, its source, and the evidence it returns — are detailed in Appendix Table A4. Together, they let liteOdyssey reason dynamically through phenotype-, gene-, and disease-level evidence during diagnostic evaluation.

### 4.3 Policy Iteration with Human Feedback (PIHF)

We developed liteOdyssey using Policy Iteration with Human Feedback (PIHF), a weight-free optimization procedure that improves a natural-language diagnostic policy without changing any model weights. PIHF is inspired by reinforcement learning with human feedback (RLHF)<sup>32</sup>: where RLHF converts human feedback into model-parameter updates, PIHF converts the same kind of feedback into revisions of a persistent, auditable policy document, optimizing the reasoning procedure rather than the model. In each iteration, a frozen reasoning-capable model is run under the current policy on development cases. The resulting benchmark scores and full reasoning process are then reviewed to locate where the reasoning broke down — a missed phenotype, premature closure on an early candidate, weak use of retrieved evidence, an unsupported gene–disease link, faulty exclusion logic, or a mis-ranked differential. For each recurrent failure, the model proposes an interpretation and a candidate revision, and a human domain expert judges which reflect clinically sound reasoning rather than artifacts of a particular benchmark. The retained feedback is written into the policy as a concrete rule that prevents that class of error; the domain expert steers the search away from brittle fixes and preserves clinically meaningful reasoning. The cycle repeats until performance stabilizes, after which the policy is frozen. Only 50 LIRICAL cases and 50 UDN cases informed this development loop; no PhenoPacket Store case was used. The policy evaluated on the public benchmarks (LIRICAL and the PhenoPacket Store) is the eight-phase workflow summarized in §4.1 and specified in full in Appendix B.

### 4.4 Benchmarks and clinical cohort

liteOdyssey was evaluated on two public phenotype-first monogenic rare-disease benchmarks and one private clinical cohort. Each public benchmark supplies a set of cases, a list of clinical features per case, and the known causal disease.

**LIRICAL** contains 370 cases across 252 diseases; approximately 45% of its causal diseases are ultra-rare by Orphanet prevalence mapping<sup>16</sup>.

**PhenoPacket Store** was evaluated over a corpus of 873 cases: a 300-case stratified sample (150 mapped and 150 unmapped), balancing mapped and unmapped rare diseases, together with a 573-case extension added to cover the broader disease distribution the PhenoPacket Store spans beyond that stratified sample. A case is designated as **mapped** when its causal disease was linked to Orphanet by our rarity-mapping pipeline, and **unmapped** when no such link was established. The unmapped label serves as a proxy for recently described, poorly indexed, or otherwise difficult-to-link rare-disease entities; it is not a claim about what the language model has or has not seen during pre-training. Across the corpus, 52.8% of causal diseases are ultra-rare.

These benchmarks were chosen because they begin at the clinical features — the point at which a real diagnostic workup begins — and because together they span a wide range of ultra-rare disease load. No PhenoPacket Store case and only 50 LIRICAL cases informed policy development, so the great majority of public cases are held out from development (see Results).

**Undiagnosed Diseases Network (UDN) cohort.** To evaluate the system under real clinical conditions, we additionally ran liteOdyssey on a private cohort of 515 patients referred to the UDN after standard clinical workup did not reach a diagnosis. Each patient had a diagnosis subsequently established by the UDN. The cohort spanned 447 distinct diagnoses across 15 geographically distributed US clinical sites (Appendix Tables A5 and A6). This cohort was run on a single closed backbone (GPT-5.3) on an institutionally approved, secure Microsoft Azure OpenAI instance; 50 cases were used for policy development, and 465 were held out. The UDN cohort was evaluated with an extended configuration of the policy: the public-benchmark policy, further developed through the same PIHF process to handle the atypical presentations and incomplete phenotype information typical of these cases. The extension adds a case-shape routing step that distinguishes typical from atypical or multi-system presentations and routes the latter through dedicated atypical-presentation reasoning, while clean, typical cases run the public policy unchanged. Only aggregate results are reported, and no patient-level material is shared. This study was approved by the Vanderbilt University Medical Center Institutional Review Board (#172005).

Stratified sampling, the rarity-mapping pipeline, and per-benchmark prevalence distributions are provided in the GitHub repository.

## 4.5 Outcomes and scoring

The primary metrics were **disease Recall@1** and **Recall@5** under a language-model judge using a standardized LLM-as-a-judge scoring prompt adapted from DeepRare<sup>12</sup> (hereafter the DeepRare-style judge). For each case, the judge received the known causal disease together with its aliases (constructed using the alias-expansion method described in DeepRare<sup>12</sup>) and the system’s ranked candidates, and determined whether the ranked output included the known causal disease and at what position; Recall@1 credits a match in the top-ranked candidate and Recall@5 a match anywhere in the top five.

Every case was additionally checked with a deterministic **exact-OMIM cross-check** that compared OMIM identifiers in the ranked output against the benchmark’s ground truth. Exact-OMIM is stricter, whereas judged-R@1 is more tolerant of synonym-level matches. For DeepSeek-family diagnostic runs, the DeepRare-style judge was itself a DeepSeek model (DeepSeek-chat), which is subject to self-preferential bias — a model can prefer its own outputs<sup>33</sup>. Those runs were therefore re-scored with the same DeepRare-style prompt using a Claude Sonnet judge. The reported numbers for DeepSeek models are from exact-OMIM and the Sonnet judge. DeepSeek-chat judge outputs are retained as sensitivity/provenance rows, not as the primary cross-model comparison surface.

For the UDN cohort, model outputs were additionally reviewed by physicians blinded to model identity. Two physicians independently scored a prespecified 120-case validation sample — one liteOdyssey and one parametric-baseline output per case (240 outputs), in randomized order — on the Bond differential-diagnosis scale (0, no close suggestion; 2, related but unlikely helpful; 3, closely related; 4, very close; 5, exact diagnosis)<sup>20</sup>. A third, also-blinded physician adjudicated the 66 output-level score disagreements (spanning 51 cases), selecting one of the prior scores or supplying an independent score. The sample was drawn under a prespecified stratified design over strict-outcome and LLM-judge-confidence strata, and scores were weighted to the full 515-case cohort by inverse selection probability. The primary comparison was the weighted mean paired Bond-score difference, evaluated with a design-based Wald test; four prespecified adjudication scenarios served as sensitivity analyses, and inter-rater agreement was summarized with the quadratic-weighted  $\kappa$ .

## 4.6 Ablation, cross-model evaluation, and reproducibility

Two contrastive evaluations isolate components of the design.

**Ablation ladder (progressively stronger configurations).** To test whether the structured workflow adds value beyond a strong prompt and access to the same information, we evaluated four cumulative configurations on the 370 LIRICAL cases, all executed on the same model (GPT-5.4) and scored under the deterministic exact-OMIM cross-check (§4.5) so that all four are graded on the same identifier-level ruler: (i) the base model prompted to diagnose from the clinical features with all biomedical tools removed, answering from internal knowledge alone using a prompting strategy adapted from prior work<sup>12</sup> — the tool-free parametric baseline used as the comparator throughout the Results, which isolates the contribution of retrieved biomedical evidence; (ii) the same model under a hardened diagnostic prompt — produced by GPT-5.5 at xhigh reasoning effort through an automatic generate–self-critique–revise–freeze procedure with adversarial review and no human authoring — and no tools; (iii) that frozen hardened prompt given access to the same public biomedical sources liteOdyssey uses, without its policy, query discipline, or adjudication; and (iv) the full liteOdyssey policy, including its engineered tools. Holding the execution model fixed isolates the effect of the prompt, the source access, and the engineered policy and tools. The hardened baseline prompt was generated autonomously by a strong frontier model (GPT-5.5, xhigh reasoning effort) and adversarially reviewed by the same model at the same reasoning effort, with no human authoring, so condition (ii) is a strong and fair prompt-only baseline rather than a weak strawman. The configurations are cumulative: each adds a layer to the one before, so the increments form a monotonic ladder and are read as such rather than as separable per-component credits (see Results; Appendix Table A1). The hardened-prompt meta-prompts and the source-access specification are provided in the Supplement.

**Cross-model evaluation.** liteOdyssey was run with the same policy and tools across different reasoning-capable models — Qwen3.6-35B-A3B, DeepSeek-V4-Flash-Preview, DeepSeek-V4-Pro-Preview, Claude Opus 4.6, and GPT-5.4 — on shared cases drawn from LIRICAL and the PhenoPacket Store (mapped, unmapped, and extension cohorts where available). For each model, the policy, tool library, workflow, prompt, and scoring were held fixed so that the model was the only variable. Recall@1, judged-R@1 verdict agreement, and per-cohort tool-call counts were recorded.

All public-benchmark runs used GPT-5.4, the open-weights Qwen3.6-35B-A3B, and the DeepSeek-V4 Flash-Preview and Pro-Preview API-routed models at high reasoning effort; the public benchmark cases will be released upon publication of the manuscript. Model versions, benchmark versions (LIRICAL release date, PhenoPacket Store snapshot date, rarity-mapping pipeline version), the tool-library version, and the judge model and prompt version are recorded in the run manifests accompanying the code release. The pipeline, tool library, benchmark construction scripts, and supplements are available at [URL TBD].

**Demo:** <https://mh-nguyen.cv/liteodyssey/>

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## Appendix A. Worked reasoning-trace cases

Public LIRICAL cases in which the full system and its own parametric baseline (the same GPT-5.4 model, differing only in the diagnostic policy and its tools) disagreed. De-identified: disease name + OMIM only; no case identifiers, iteration names, or internal tool labels. Myhre syndrome is summarized in the main text (see Results) and traced in full below as a worked example, along five dimensions a clinician would inspect — the phenotypes that seeded the hypothesis, the candidates considered, the evidence that supported or weakened each, where uncertainty remained, and what a reviewer could inspect — to show concretely how the policy and tools reach a diagnosis the unaided model misses. The remaining cases are documented more briefly.

### Rescues — where the policy and tools earn the diagnosis

**Myhre syndrome (OMIM 139210; causal gene SMAD4).** *(also in the Results; parsed transcript outputs in Appendix Table A7.)* *Phenotypes that seeded the hypothesis:* a sclerosing, hyperostotic skeletal presentation with a distinctive signature — enlarged vertebral pedicles and a thickened calvaria — alongside brachydactyly, joint stiffness, and conductive hearing loss. *Candidates considered:* the parametric baseline pattern-matched this presentation to a basket of hyperostotic and filamin-related dysplasias, ranking Lenz–Majewski hyperostotic dwarfism first, with frontometaphyseal dysplasia, otopalatodigital-spectrum disorders, and Melnick–Needles syndrome below it, and never reached the correct disease. The full system instead surfaced SMAD4 and adjudicated it against its nearest competitor, NFIX (Marshall–Smith syndrome). *Evidence that supported or weakened each candidate:* the full system’s phenotype-to-gene retrieval quantified SMAD4 as the top-scoring gene (information-content score 1.1137 across 11 matched terms); an independent disease-level lookup returned Myhre syndrome as the top hit; and gene–disease validity and population-constraint evidence confirmed SMAD4 as a Definitive, constraint-intolerant dominant gene. In adjudication, only Myhre syndrome accounted for the thickened calvaria, enlarged vertebral pedicles, brachydactyly, and joint stiffness, which NFIX-related disease does not. *Where uncertainty remained:* little — the system reported HIGH confidence, the specific skeletal signature anchoring SMAD4 where the unaided model had defaulted to the more familiar hyperostotic disorders. *What a reviewer could inspect:* a quantified evidence chain assembled across 29 tool calls (phenotype-to-gene ranking, disease lookup, gene–disease validity, population constraint), each step traceable to its public source; Myhre syndrome was ranked first. The parsed side-by-side outputs of both configurations are reported in Appendix Table A7.

**Cornelia de Lange syndrome type 3 (OMIM 610759).** The baseline correctly recognized the Cornelia de Lange syndrome family but defaulted to the canonical gene of a different subtype (NIPBL, type 1), which the diagnosis-level judge scores as a miss. The full system’s trace shows a focused re-query on the discriminating distal-limb and periocular features — curly eyelashes, proximal thumbs, short fourth and fifth metacarpals — holding the correct subtype gene above the canonical one; it ranked Cornelia de Lange syndrome type 3 first. Retrieved, case-specific evidence overrode a strong but wrong textbook default.

### A shared miss, and a regression

**Acromicric dysplasia (OMIM 102370).** Both configurations missed. The baseline produced a confident but incorrect list of chromatin-related syndromes; the full system anchored on severe

primordial growth failure and locked its search onto a primordial-dwarfism family, never surfacing the correct acromelic dysplasia. The difference is that the full system’s trace records the wrong anchor explicitly and the system reported lower confidence on the case, whereas the baseline gave no such signal.

**Muenke syndrome (OMIM 602849).** Here the policy and tools cost a correct answer. The parametric baseline ranked Muenke syndrome first from its textbook prior; the full system, anchoring on bicoronal synostosis and enumerating the rarer craniosynostosis genes, over-elaborated past the common diagnosis and dropped it out of the top five.

## Appendix B. Diagnostic policy: the eight-phase reasoning workflow

The policy produced by PIHF (§4.3) is an eight-phase clinical reasoning workflow, executed end-to-end by a single reasoning-capable model. The workflow is structured but not strictly linear: the system can revisit earlier phases when new evidence changes the working differential or when uncertainty remains high. Each phase specifies which biomedical tools (§4.2) it draws on, so the phases below also indicate how tool-derived evidence enters the diagnostic process.

**Phase 0 — Pattern recognition.** The system reviews the HPO term list to form an initial hypothesis about the likely disease family, anchored on the most specific features.

**Phase 1 — Phenotype-based candidate gene generation.** The system queries the phenotype-to-gene tool, with rare and specific terms weighted more heavily than common terms by information content. The output is an initial gene shortlist based on similarity between the patient’s clinical features and known gene-phenotype associations.

**Phase 2 — Evidence triage.** Each gene on the shortlist is evaluated for curated gene-disease validity. Genes with definitive, strong, or moderate evidence are prioritized for deeper evaluation; genes with limited or disputed evidence are deprioritized unless they remain uniquely consistent with the phenotype.

**Phase 3 — Deep investigation.** For prioritized gene candidates, the system retrieves disease-level evidence — inheritance, age of onset, associated clinical features, related syndromes — and builds a case for or against each candidate by comparing the disease’s expected presentation against the patient’s observed features.

**Phase 4 — Confidence assessment.** The system assesses whether the top candidate explains the full phenotype, whether any feature cluster remains unexplained, and whether a related syndrome could provide a better fit.

**Phase 5 — Corrective search.** When confidence is low or a clinical feature cluster remains unexplained, the system performs targeted literature and disease-family searches for recently described or poorly indexed candidates. This step addresses cases in which the initial ranking missed the correct answer because the disease was too new or too rare to appear prominently.

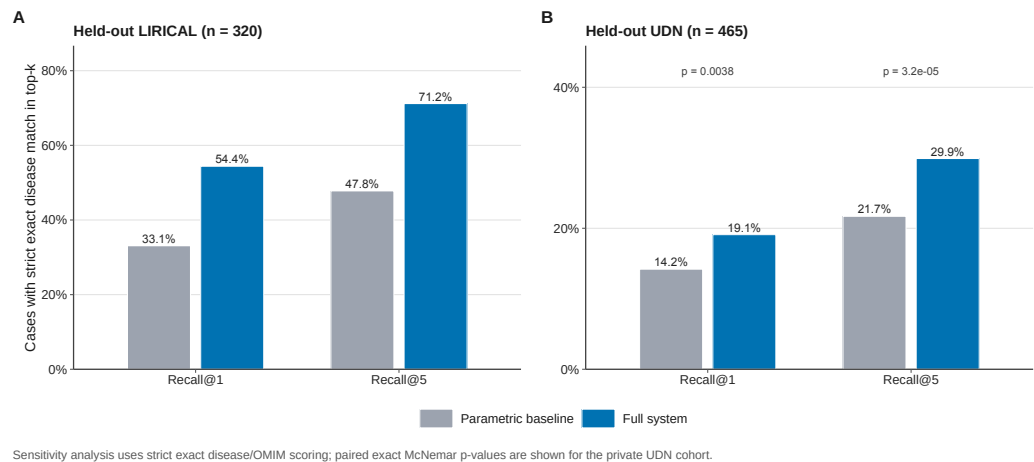
**Phase 6 — Reflective adjudication.** Before finalizing the differential, the system re-examines the leading candidate against its competitors, including those with greater phenotypic specificity or less common but more clinically consistent presentations. This step targets a recurrent error mode: when two related candidates are plausible, a model may otherwise favor the more common or better-annotated disorder over a rarer but more phenotypically specific alternative. The system asks three pre-defined questions — whether the leading diagnosis explains all major feature clusters, whether

a competing candidate provides a more specific phenotypic fit, and whether targeted literature or disease-family search could alter the ranking.

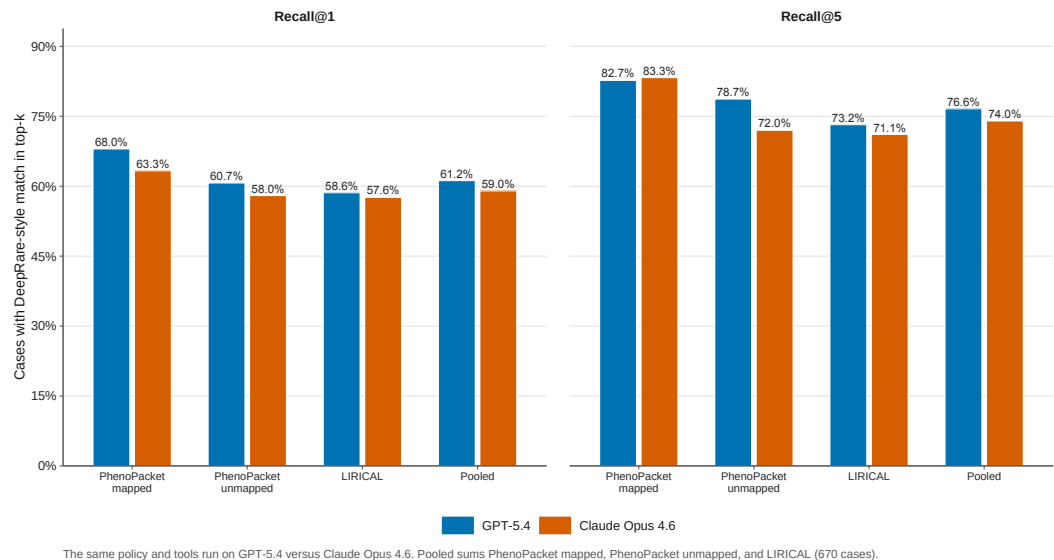
**Phase 7 — Final output.** The system produces a ranked top-five differential diagnosis. For each candidate, it reports the disease, associated gene, supporting evidence, confidence classification, and the rationale for its position in the ranking.

Because the model executes these phases in sequence, the workflow yields a single continuous reasoning process — every tool call, intermediate ranking, and reflective re-check recorded in order. This record provides the basis for clinical auditability (see Results): a reviewing clinician can follow, step by step, how the differential was constructed and challenge any inference that lacks retrieved support.

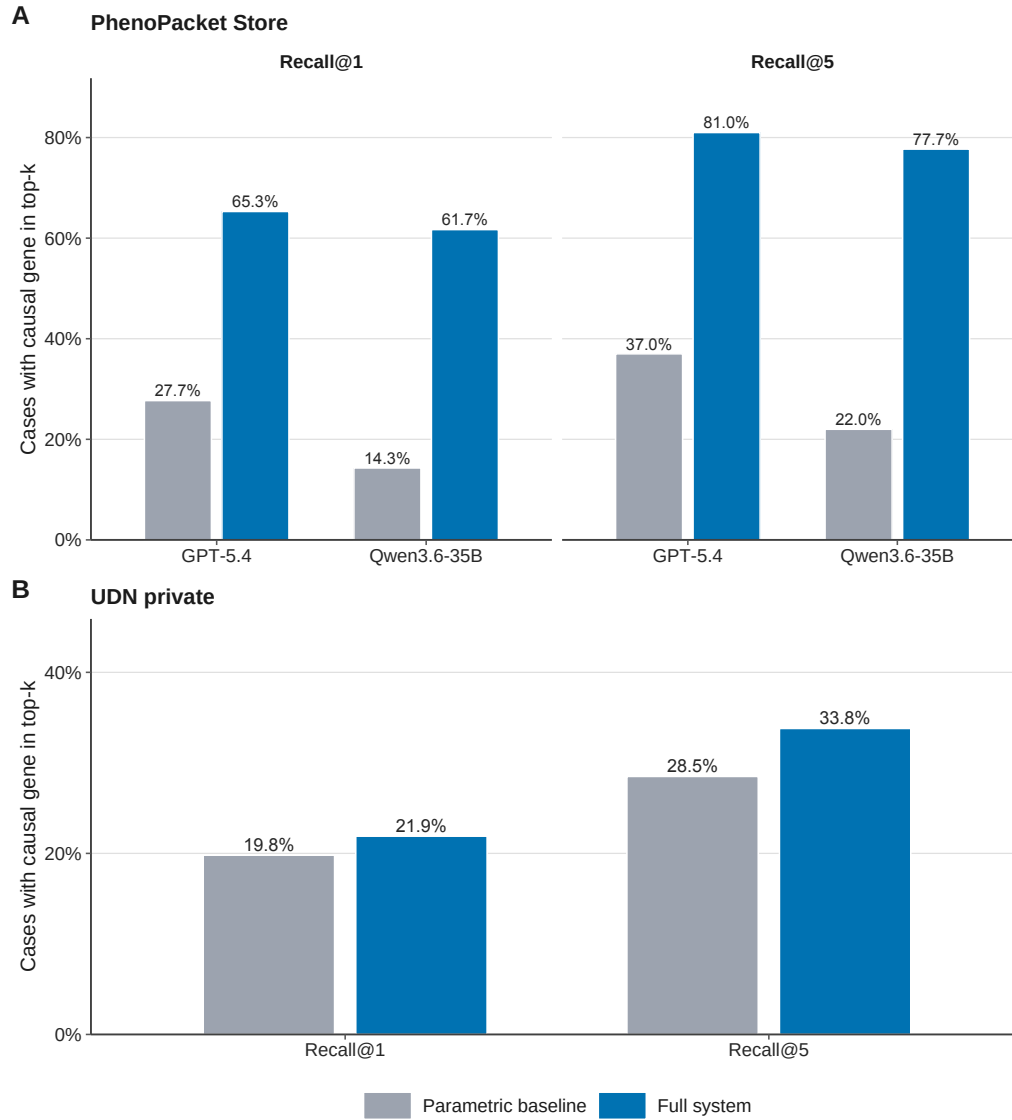
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Appendix Figure A1. Exact-OMIM disjoint sensitivity — the development-excluded analysis (Figure 5) re-scored under the stricter deterministic exact-OMIM cross-check.



Appendix Figure A2. Full-system R@1 under the same policy on Claude Opus 4.6 versus GPT-5.4 across 670 cases.



Gene-level (causal-gene) ranking: full system vs parametric baseline. PhenoPacket Store pooled (300; GPT-5.4 and Qwen3.6-35B) a

Appendix Figure A3. Gene-level prediction — causal-gene ranking, full system versus parametric baseline. On the PhenoPacket Store (GPT-5.4), the full system ranked the causal gene first in 65.3% of cases and within the top five in 81.0%, versus 27.7% and 37.0% for the parametric baseline (+37.6 and +44.0 points); the open-weights Qwen3.6 showed the same pattern (61.7% versus 14.3% at Recall@1). On the 515-case UDN cohort (GPT-5.3), the gain was attenuated and concentrated at Recall@5 (33.8% versus 28.5%; paired McNemar  $p = 0.0049$  after Bonferroni correction), while the Recall@1 difference was not significant (21.9% versus 19.8%;  $p = 0.21$ ). Gene-level accuracy is scored only where causal-gene ground truth exists (the PhenoPacket Store and UDN; LIRICAL is defined at the OMIM-disease level and does not include gene labels).

**Appendix Table A1. Progressively stronger configurations on LIRICAL (exact-OMIM cross-check, 370 cases), all executed on the same model (GPT-5.4). Each configuration adds a layer to the one above; the configurations are cumulative and their individual contributions are not separable.**

| Configuration                                                   | Recall@1 | Recall@5 |
|-----------------------------------------------------------------|----------|----------|
| Base model (clinical features only)                             | 34.3%    | 48.6%    |
| + hardened, autonomously generated diagnostic prompt (no tools) | 36.8%    | 52.7%    |
| + access to the same public biomedical sources                  | 44.9%    | 61.9%    |
| Full liteOdyssey policy and engineered tools                    | 55.9%    | 72.7%    |

**Appendix Table A2. Disease distribution across development and held-out evaluation cases.**

| Cohort                                         | Cases | Unique diseases | Development            | Held out                 |
|------------------------------------------------|-------|-----------------|------------------------|--------------------------|
| LIRICAL                                        | 370   | 252             | 50 cases (43 diseases) | 320 cases                |
| PhenoPacket Store — mapped                     | 150   | 125             | 0                      | 150                      |
| PhenoPacket Store — unmapped                   | 150   | 139             | 0                      | 150                      |
| PhenoPacket Store — extension                  | 573   | 297             | 0                      | 573                      |
| Public total (deduplicated union) <sup>1</sup> | 1,243 | 722             | 50 cases (43 diseases) | 1,193 cases              |
| UDN (private)                                  | 515   | 447             | 50 cases (50 diseases) | 465 cases (407 diseases) |

| Cohort                     | Endpoint | Workflow    | N   | Judge           | Exact OMIM      | Judge - exact, pp |
|----------------------------|----------|-------------|-----|-----------------|-----------------|-------------------|
| Held-out LIRICAL (n = 320) | Recall@1 | Full system | 320 | 184/320 (57.5%) | 174/320 (54.4%) | +3.1              |
|                            |          | Parametric  | 320 | 109/320 (34.1%) | 106/320 (33.1%) | +1.0              |
|                            | Recall@5 | Full system | 320 | 230/320 (71.9%) | 228/320 (71.2%) | +0.7              |
|                            |          | Parametric  | 320 | 163/320 (50.9%) | 153/320 (47.8%) | +3.1              |
| Held-out UDN (n = 465)     | Recall@1 | Full system | 465 | 89/465 (19.1%)  | 89/465 (19.1%)  | +0.0              |
|                            |          | Parametric  | 465 | 72/465 (15.5%)  | 66/465 (14.2%)  | +1.3              |
|                            | Recall@5 | Full system | 465 | 139/465 (29.9%) | 139/465 (29.9%) | +0.0              |
|                            |          | Parametric  | 465 | 110/465 (23.7%) | 101/465 (21.7%) | +2.0              |

Appendix Table A3. Judge-style versus exact-OMIM scoring, development-excluded rows (the 320 held-out LIRICAL and 465 held-out UDN cases).

**Appendix Table A4. liteOdyssey biomedical tool library — the clinical question each tool answers, its source, and the evidence it returns (the five phenotype-first knowledge**

tools used on the public benchmarks; the two variant-mode components are described in §4.2).

| Tool             | Clinical question                                                            | Source                                                                             | Evidence returned                                                                           |
|------------------|------------------------------------------------------------------------------|------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|
| dx-monarch       | Which genes match the patient’s phenotypes?                                  | Monarch Initiative                                                                 | Information-content-weighted candidate gene list                                            |
| dx-omim          | What diseases does this gene cause / which genes are in this disease family? | OMIM (local files: genemap2, morbidmap, mimTitles; HPO annotations phenotype.hpoa) | Disease records (inheritance, age of onset), keyword search, and disease-series enumeration |
| dx-gene-evidence | Is the gene intolerant of loss-of-function / missense variation?             | gnomAD (pLI, LOEUF, missense_z) + OMIM                                             | Per-gene constraint metrics + OMIM diseases                                                 |
| dx-clingen       | How strong is the curated gene-disease link / dosage sensitivity?            | ClinGen (local reference CSVs)                                                     | Gene-disease validity + haploinsufficiency classification                                   |
| dx-pubmed        | Are there relevant case reports for this presentation?                       | PubMed (NCBI E-utilities)                                                          | Relevant abstracts                                                                          |

**Appendix Table A5. UDN cohort composition.**

| Property                           | Value                                                      |
|------------------------------------|------------------------------------------------------------|
| Patients                           | 515 (50 development, 465 held out)                         |
| Distinct diagnoses                 | 447 (407 among the held-out patients)                      |
| Diagnoses seen in a single patient | 411 of 447                                                 |
| Most frequent diagnosis            | ReNU syndrome, 25 patients                                 |
| Clinical sites                     | 15 geographically distributed US sites (Appendix Table A6) |

**Appendix Table A6. UDN geographic distribution across 15 US clinical sites (all 515 patients); the four smallest sites (1–2 patients each) are grouped.**

| Clinical site                        | Patients | %    |
|--------------------------------------|----------|------|
| NIH Undiagnosed Diseases Program     | 110      | 21.4 |
| Vanderbilt University Medical Center | 70       | 13.6 |
| Duke University Medical Center       | 67       | 13.0 |
| UDN Central                          | 64       | 12.4 |
| Baylor College of Medicine           | 60       | 11.7 |
| Stanford University                  | 46       | 8.9  |

| Clinical site                                            | Patients | %   |
|----------------------------------------------------------|----------|-----|
| University of Washington and Seattle Children’s Hospital | 29       | 5.6 |
| University of California, Los Angeles                    | 20       | 3.9 |
| Harvard-affiliated Hospitals (BCH, BWH, MGH)             | 19       | 3.7 |
| Washington University in St. Louis                       | 18       | 3.5 |
| University of Miami                                      | 7        | 1.4 |
| Four additional sites (1–2 patients each)                | 5        | 1.0 |

**Appendix Table A7. Myhre syndrome (OMIM 139210, SMAD4) — parsed outputs from the actual run transcripts, full system versus parametric baseline (same GPT-5.4 model).**

| Element                           | Full system (policy + tools)                                                                                                                                                            | Parametric baseline (same model, no tools)                                                                                                  |
|-----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| Top-ranked disease                | Myhre syndrome (OMIM 139210) — correct                                                                                                                                                  | Lenz–Majewski hyperostotic dwarfism (OMIM 151050) — incorrect                                                                               |
| Reported confidence               | HIGH                                                                                                                                                                                    | MODERATE                                                                                                                                    |
| Top-5 differential (gene)         | Myhre (SMAD4) ·<br>Marshall–Smith (NFIX) ·<br>Kabuki 1 (KMT2D) ·<br>proximal symphalangism (NOG) · osteopathia striata with cranial sclerosis (AMER1)                                   | Lenz–Majewski (PTDSS1) ·<br>frontometaphyseal dysplasia 1 (FLNA) · KBG (ANKRD11) ·<br>otopalatodigital 1 (FLNA) ·<br>Melnick–Needles (FLNA) |
| Correct disease (SMAD4) in output | Rank 1                                                                                                                                                                                  | Absent from top 5                                                                                                                           |
| Tool calls                        | 29 (phenotype-to-gene, disease lookup, gene–disease validity, population constraint)                                                                                                    | none                                                                                                                                        |
| Key evidence                      | SMAD4 the top phenotype-to-gene hit (information-content score 1.1137, 11 matched terms); OMIM returned Myhre as top disease; ClinGen Definitive, autosomal dominant; gnomAD constraint | pattern-match from model prior to the hyperostotic / filamin family                                                                         |
| Adjudication                      | Myhre (SMAD4) held over Marshall–Smith (NFIX): only Myhre explains the thickened calvaria, enlarged vertebral pedicles, brachydactyly, and joint stiffness                              | none                                                                                                                                        |